Jiya Sinha

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Research Interests: Machine Learning · Deep Learning · Computer Vision · Audio Processing

Education

Indian Institute of Science Education and Research,Bhopal 2022- 2027
BS-MS Data Science and Engineering *CPI (7^{th} Sem): 8.5/10*

Laxman Public School, New Delhi 2020-2022
Class XII (CBSE): 94%

Mount Carmel School, Chandigarh 2016-2020
Class X (CBSE): 96.33%

Relevant Courses

Deep Learning, Machine Learning, Natural Language Processing, Computer Vision, 3D Deep Learning and Applications, Artificial Intelligence, Machine Learning with graphs, Applied Accelerated Artificial Intelligence, Data Structures and Algorithms, Databases, Probability and Statistics, Linear Algebra, Calculus, Signals and Systems

Publications

Sinha, J., Bhattacharya, P., & Agarwal, A. (2025). Gesture recognition for emergencies: Dataset and cross-condition analysis. *2025 IEEE 19th International Conference on Automatic Face and Gesture Recognition (FG)*, 1–6. https://doi.org/10.1109/FG61629.2025.11099316

Thesis

- **Audio SceneFake Detection** *Jan 2026 – Apr 2026*
  Bachelor's Thesis
  Supervisor: Dr. Akshay Agarwal, Assistant Professor, IISER Bhopal
  https://drive.google.com/file/d/1-QDfykf2f5fAxdnW_OipbtKi8FQ9XUjl/view?usp=sharing
  - ∗ Investigated acoustic scene manipulation (SceneFake) and developed a 15.42-hour reality-aware audio dataset collected under natural recording conditions.
  - ∗ Studied generalization under distribution shifts; demonstrated importance of limited real-world data.
  - ∗ Evaluated SOTA audio deepfake detection models under cross-dataset and OOD settings.

Internships

- **Research Intern** *September 2024 – January 2025*
  Trustworthy BiometraVision Lab, IISER Bhopal *Certificate*
  - ∗ Created a 1,000+ video gesture recognition dataset across diverse environments/devices, enabling robust detection of five dynamic emergency gestures by Indian participants.
  - ∗ Implemented and benchmarked SOTA deep learning models (ViViT, VGG-LSTM, MobileNetV2-LSTM), analyzing performance under varying lighting and device quality.
  - ∗ Co-authored a peer-reviewed paper at FG2025.
- **Summer Intern** *May 2025 – Aug 2025*
  Advanced Signal and Image Processing Lab, IISER Bhopal
  Code: https://github.com/sinhajiya/PerceptGAN
  - ∗ Designed deep learning models for colorizing near-infrared (NIR) images to RGB.
  - ∗ Implemented GAN-based approaches with multiple generator architectures and custom loss functions.
  - ∗ Optimized trade-off between perceptual quality and computational efficiency for real-world deployment.

Personal Projects

- **Graph-Based Makeup Transfer using GCNs** *PyTorch Geometric, GCNs, Computer Vision*
  Code: https://github.com/sinhajiya/graphglow.git
  * Developed a facial landmark graph model combining histogram matching with GCN-based refinement for structure-aware makeup transfer.
  * Designed cross-face and intra-face message passing for stable color propagation across facial regions (e.g., lips, eyes).
  * Trained on the MT dataset; evaluated via a 54-user study (avg. score: 4.9/10), highlighting scope for improvement.
- **Face Recognition** *Python, Scikit-learn, Computer Vision*
  Code: https://github.com/sinhajiya/Face_Recognition_from_Features_using_Yale_Face_Database.git
  * Built a face recognition pipeline using the Yale Face Database with hand-crafted feature extraction and classical ML models.
  * Implemented preprocessing, feature engineering, and model evaluation using traditional computer vision techniques.

Skills & Tools

- **Programming:** Python
- **Machine Learning & Deep Learning:** scikit-learn, PyTorch, PyTorch Geometric
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, OpenCV

Personal Details and Links

- **Email-id** jiya22@iiserb.ac.in
- **Personal Email-id** jiyasinha2004b@gmail.com
- **LinkedIn:** www.linkedin.com/in/jiyasinha
- **GitHub:** https://github.com/sinhajiya